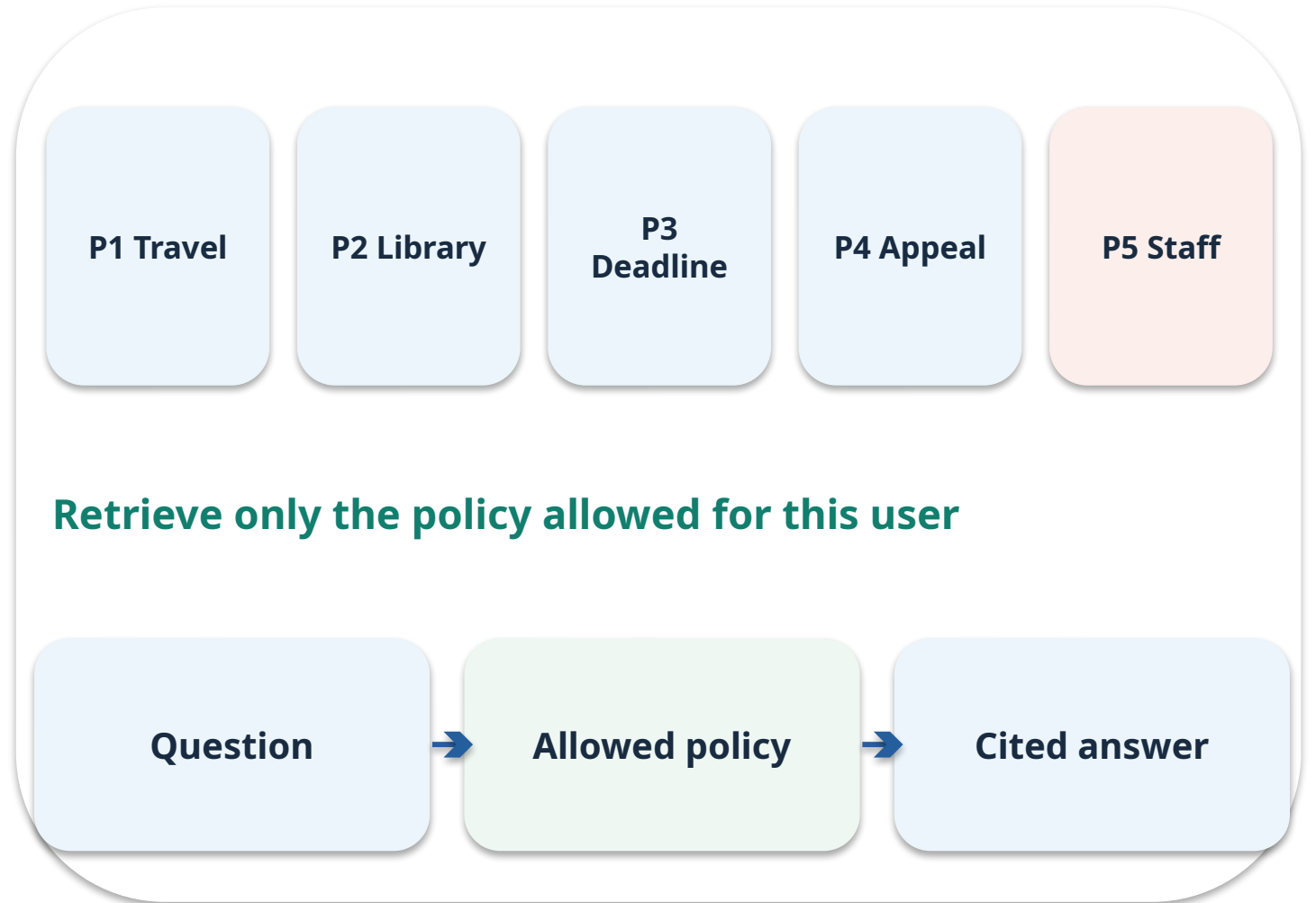


Week 06 · Finding useful information

OPEN / WEEK 06

The library-renewal search

A learner asks “Can I renew my book?” Policy P2 says “library loans may be extended once.” Which passage should search find?



First vote: what would you do?

QUESTION / WEEK 06

A learner asks “Can I renew my book?” Policy P2 says “library loans may be extended once.” Which passage should search find?

Choose an initial answer and name one piece of evidence you need.

VOTE FIRST • GIVE A REASON

A Accept the first plausible result

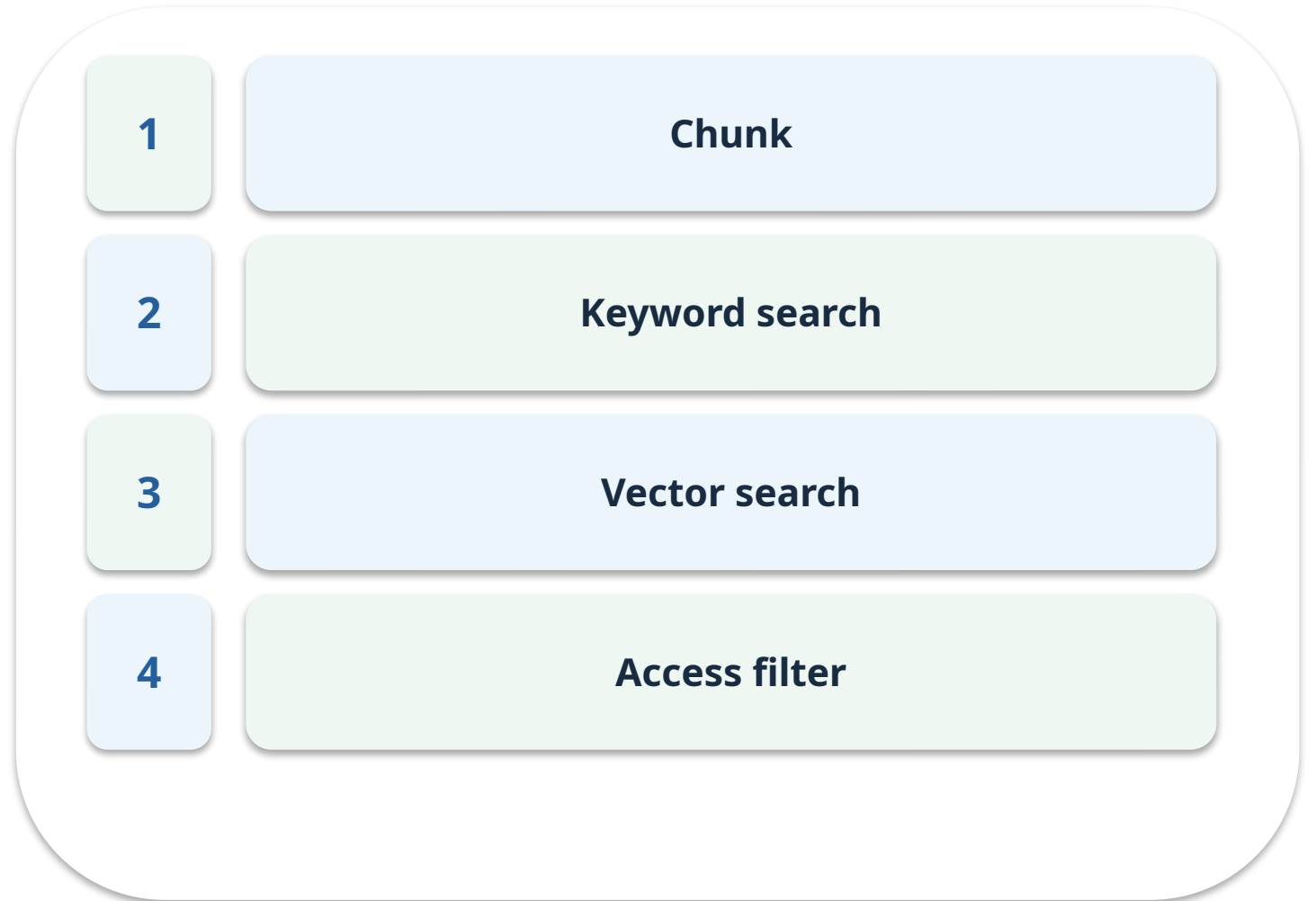
B Check the source or measurement

Which source or test could settle this?

The journey through today's case

OPEN / WEEK 06

We will predict, connect an analogy, trace a mechanism, work an example, then explain a decision.



1. Predict · Chunk

QUESTION / WEEK 06

Why preserve the source ID and version?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A They let a learner verify the answer in the correct policy edition.

B A chunk needs only its text.

Which source or test could settle this?

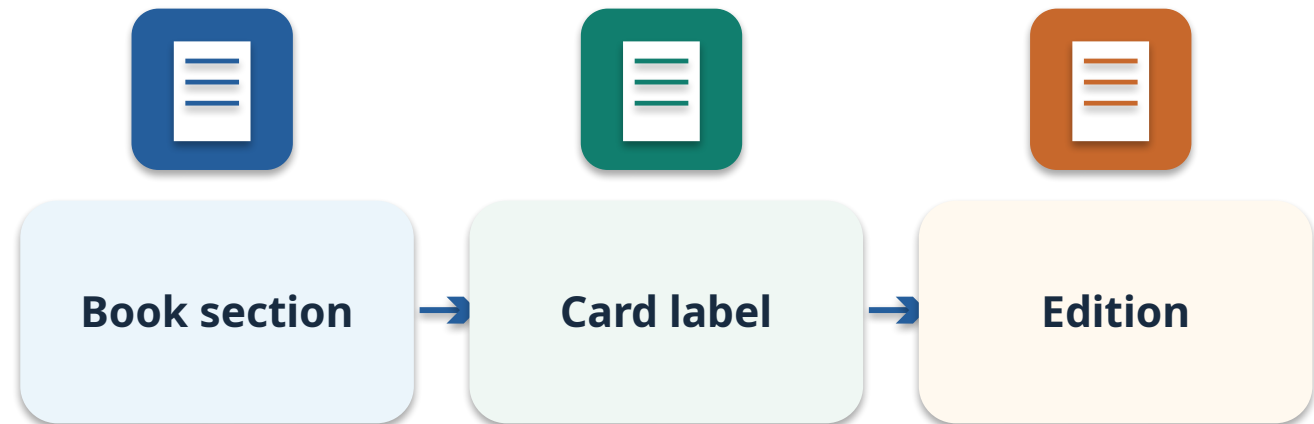
Chunk · a familiar picture

ANALOGY / WEEK 06

A long textbook gets labelled index cards for individual sections.

What matches our case, and where does the analogy stop?

A long textbook gets labelled index cards for individual sections.



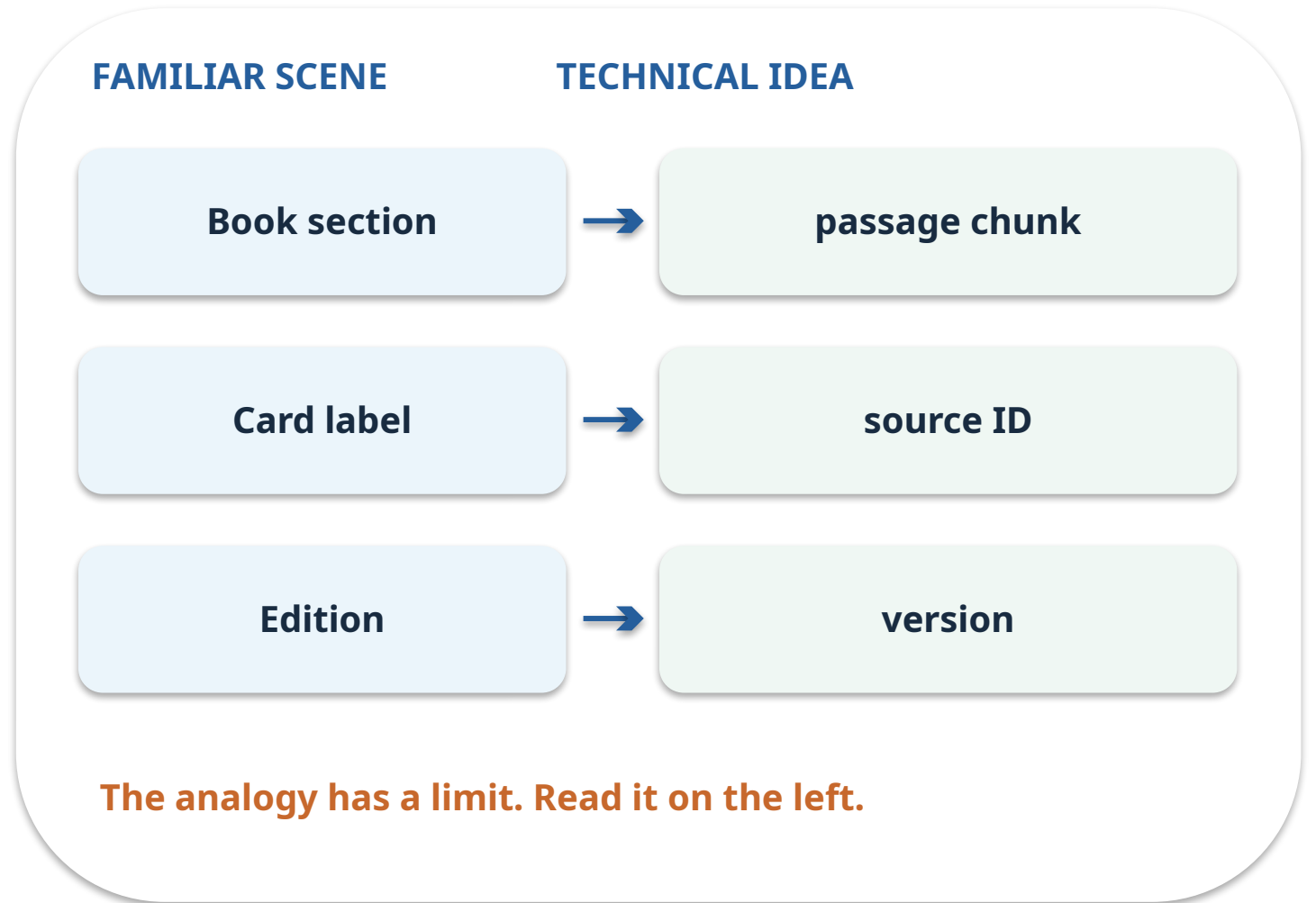
What matches the technical process?

Chunk · map the parts

MAPPING / WEEK 06

Match each everyday part to the technical part.

Analogy limit: Cutting at arbitrary length can separate a condition from its rule.



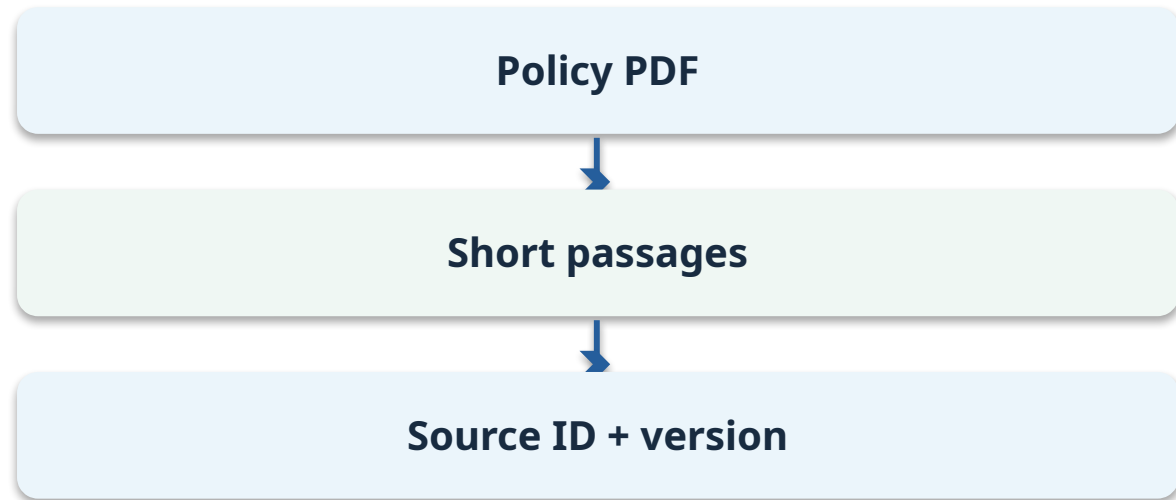
Chunk · how it works

MECHANISM / WEEK 06

A short passage from a larger source; keep its document ID and version.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Chunk · work one case

WORKED / WEEK 06

P2 becomes a short passage with ID P2 and version; keep “once” in the same passage.

Point to the evidence before giving your conclusion.

GIVEN

P2 becomes a short passage with ID P2 and version; keep “once” in the same passage.

QUESTION

Why preserve the source ID and version?

SUPPORTED ANSWER

They let a learner verify the answer in the correct policy edition.

Chunk · answer and explain

REVEAL / WEEK 06

They let a learner verify the answer in the correct policy edition.

Tempting mistake: A chunk needs only its text.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A chunk needs only its text.

SUPPORTED EXPLANATION

They let a learner verify the answer in the correct policy edition.

Cutting at arbitrary length can separate a condition from its rule.

2. Predict · Keyword search

QUESTION / WEEK 06

What explains this miss?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A Different words express a similar intent.

B No exact match proves no relevant policy exists.

Which source or test could settle this?

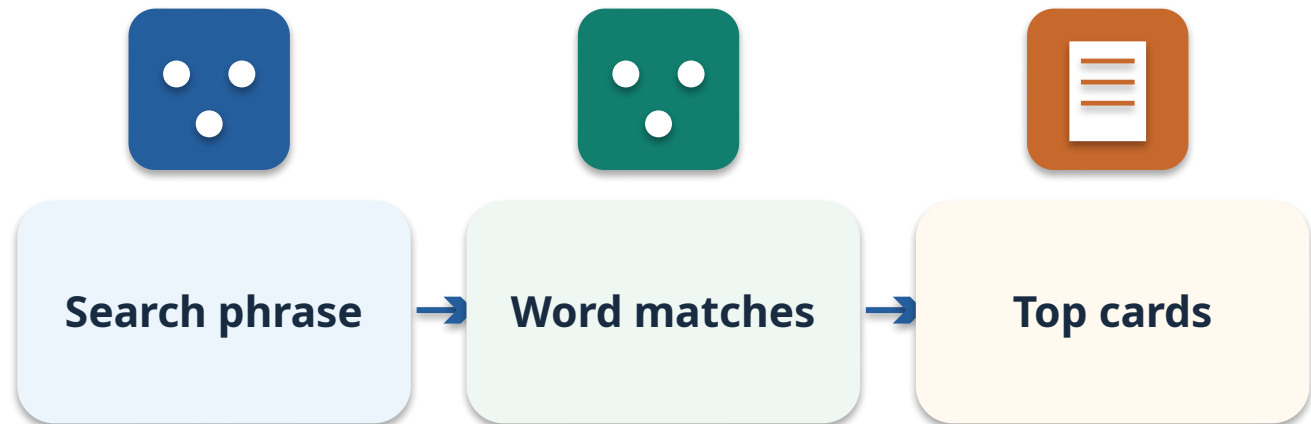
Keyword search · a familiar picture

ANALOGY / WEEK 06

An index finds pages that contain the exact words you entered.

What matches our case, and where does the analogy stop?

An index finds pages that contain the exact words you entered.



What matches the technical process?

Keyword search · map the parts

MAPPING / WEEK 06

Match each everyday part to the technical part.

Analogy limit: Synonyms can make a relevant passage score poorly.

FAMILIAR SCENE

Search phrase

Word matches

Top cards

TECHNICAL IDEA

query tokens

overlap score

ranked passages

The analogy has a limit. Read it on the left.

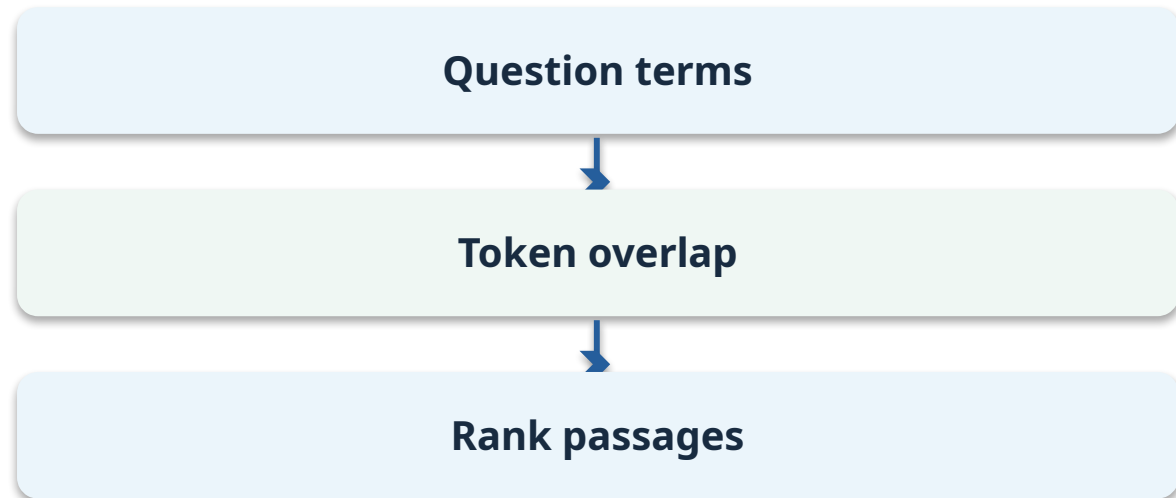
Keyword search · how it works

MECHANISM / WEEK 06

Matches terms in a query and passage; useful when wording overlaps.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Keyword search · work one case

WORKED / WEEK 06

“renew” does not exactly match P2’s “extend”; a keyword-only search may miss it.

Point to the evidence before giving your conclusion.

GIVEN

“renew” does not exactly match P2’s “extend”; a keyword-only search may miss it.

QUESTION

What explains this miss?

SUPPORTED ANSWER

Different words express a similar intent.

Keyword search · answer and explain

REVEAL / WEEK 06

Different words express a similar intent.

Tempting mistake: No exact match proves no relevant policy exists.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

No exact match proves no relevant policy exists.

SUPPORTED EXPLANATION

Different words express a similar intent.

Synonyms can make a relevant passage score poorly.

3. Predict · Vector search

QUESTION / WEEK 06

Which passage ranks nearer to q in this toy calculation?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A P2, with the larger cosine similarity.

B An embedding is an authoritative answer.

Which source or test could settle this?

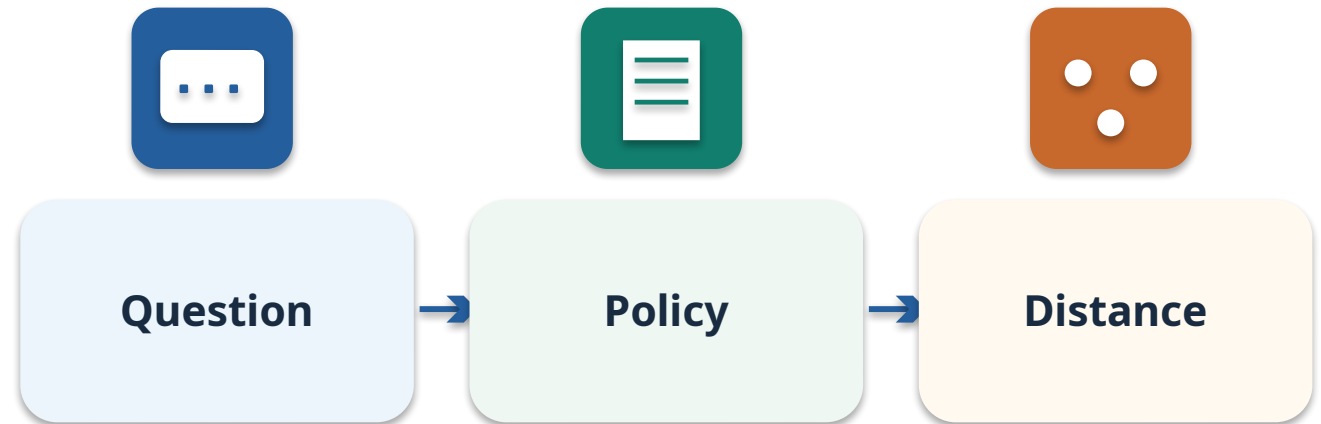
Vector search · a familiar picture

ANALOGY / WEEK 06

A map places related meanings near one another.

What matches our case, and where does the analogy stop?

A map places related meanings near one another.



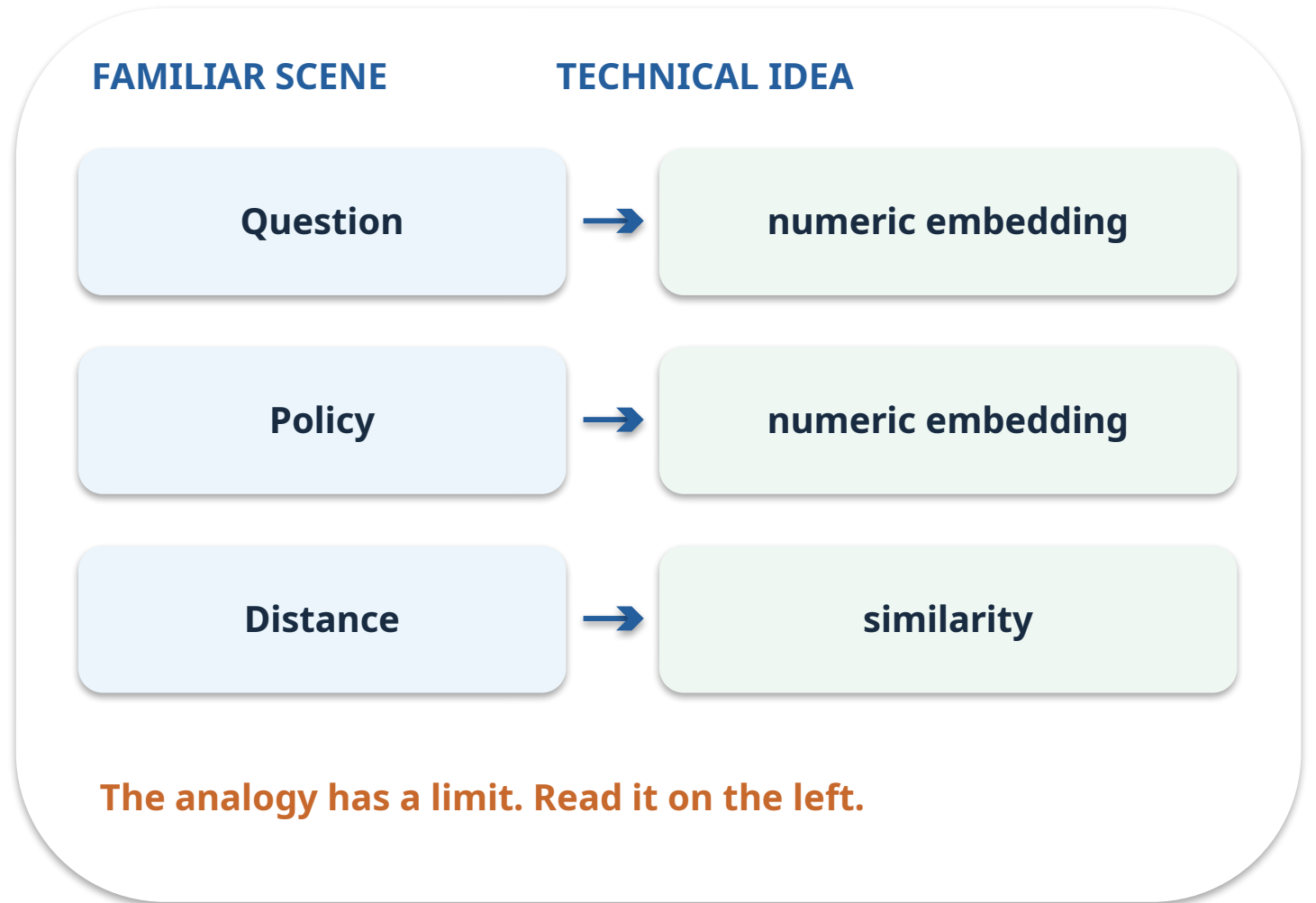
What matches the technical process?

Vector search · map the parts

MAPPING / WEEK 06

Match each everyday part to the technical part.

Analogy limit: Similar vectors can retrieve the wrong passage; check text and permissions.



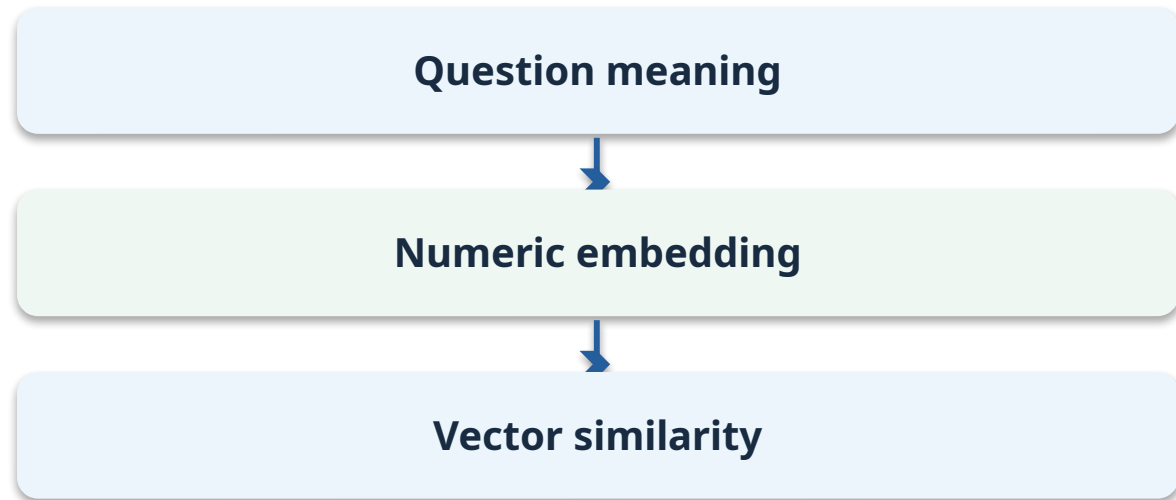
Vector search · how it works

MECHANISM / WEEK 06

Compares numerical representations of text; a production embedding model can capture semantic similarity.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Vector search · work one case

WORKED / WEEK 06

Toy vectors $q=[1,0]$, $P2=[1,1]$, $P3=[0,1]$: cosine scores about .707 and 0.

Point to the evidence before giving your conclusion.

GIVEN

Toy vectors $q=[1,0]$, $P2=[1,1]$, $P3=[0,1]$: cosine scores about .707 and 0.

QUESTION

Which passage ranks nearer to q in this toy calculation?

SUPPORTED ANSWER

P2, with the larger cosine similarity.

Vector search · answer and explain

REVEAL / WEEK 06

P2, with the larger cosine similarity.

Tempting mistake: An embedding is an authoritative answer.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

An embedding is an authoritative answer.

SUPPORTED EXPLANATION

P2, with the larger cosine similarity.

Similar vectors can retrieve the wrong passage; check text and permissions.

4. Predict · Access filter

QUESTION / WEEK 06

When must the access filter apply?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A Before retrieval results are passed into an answer.

B A relevant staff policy may be shown if cited.

Which source or test could settle this?

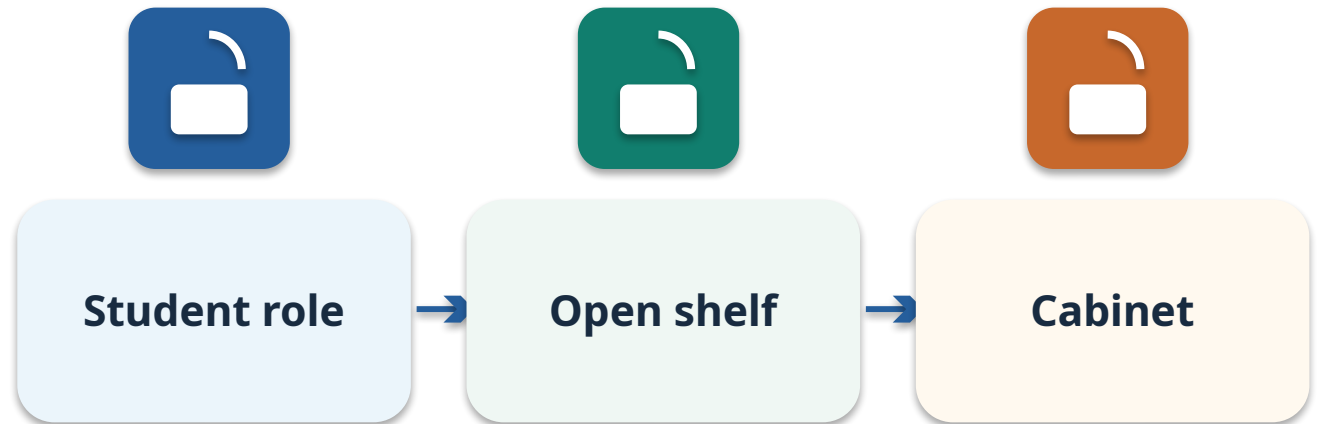
Access filter · a familiar picture

ANALOGY / WEEK 06

A library card allows entry to some shelves but not staff cabinets.

What matches our case, and where does the analogy stop?

A library card allows entry to some shelves but not staff cabinets.



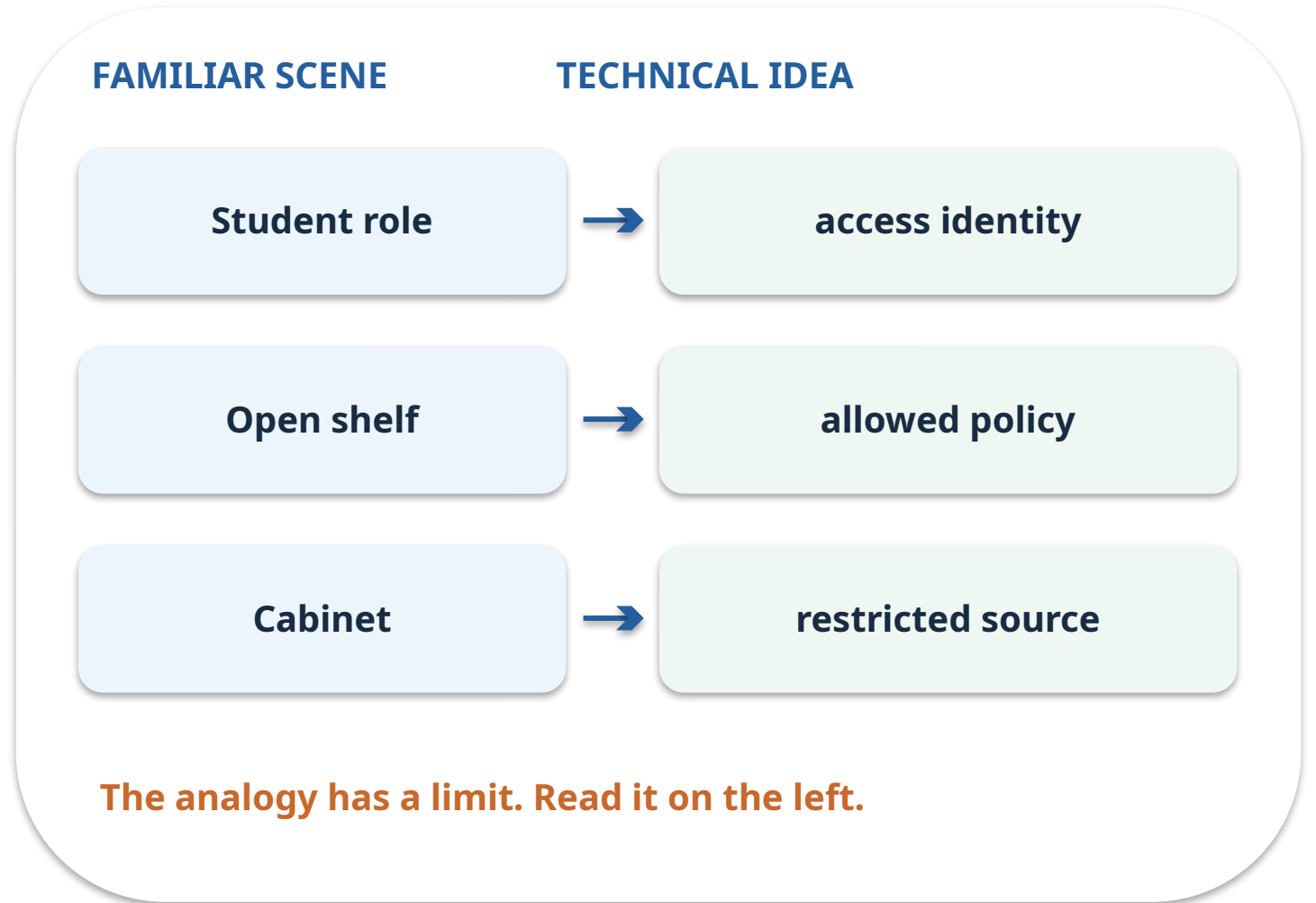
What matches the technical process?

Access filter · map the parts

MAPPING / WEEK 06

Match each everyday part to the technical part.

Analogy limit: Filtering after generation may leak details already inserted into context.



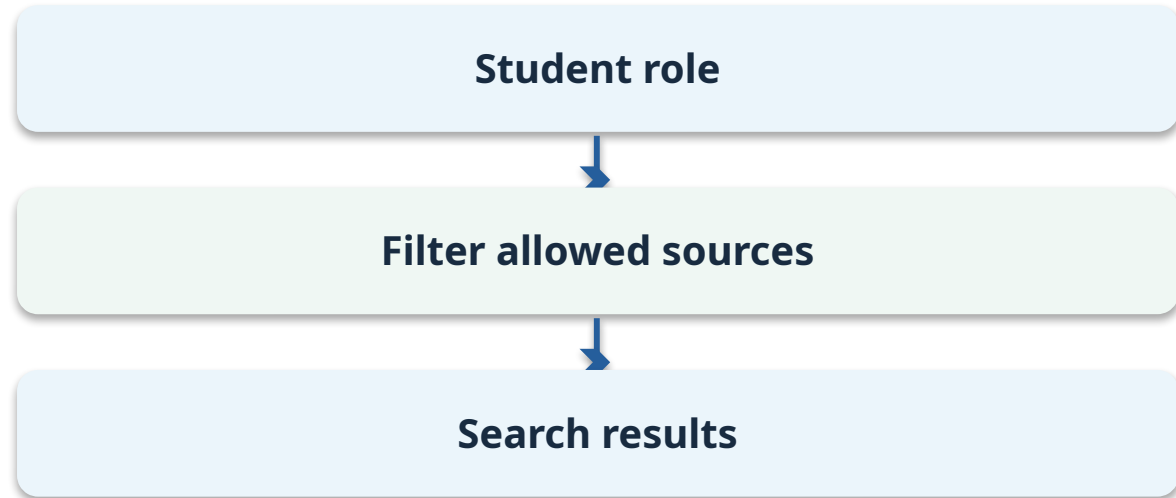
Access filter · how it works

MECHANISM / WEEK 06

Only retrieve passages that the current role is allowed to read.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Access filter · work one case

WORKED / WEEK 06

Filter staff-only P5 before ranking; never supply it to the student answer stage.

Point to the evidence before giving your conclusion.

GIVEN

Filter staff-only P5 before ranking; never supply it to the student answer stage.

QUESTION

When must the access filter apply?

SUPPORTED ANSWER

Before retrieval results are passed into an answer.

Access filter · answer and explain

REVEAL / WEEK 06

Before retrieval results are passed into an answer.

Tempting mistake: A relevant staff policy may be shown if cited.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A relevant staff policy may be shown if cited.

SUPPORTED EXPLANATION

Before retrieval results are passed into an answer.

Filtering after generation may leak details already inserted into context.

Evidence trail: input → result

WORKED / WEEK 06

Question uses the phrase renew a book.

Exact overlap may miss P2; a simple synonym mapping improves this case but can create other misses.

GIVEN

P2 says library loans may be extended once.

QUESTION

Question uses the phrase renew a book.

SUPPORTED ANSWER

Exact overlap may miss P2; a simple synonym mapping improves this case but can create other misses.

Notebook: run the prepared example

PRACTICE / WEEK 06

Compare token overlap and the synonym proxy on a fixed corpus.

Read the original output before editing.

1

Create a five-policy corpus with access labels

2

Search ten questions with token overlap

3

Use the supplied synonym expansion as a meaning-aware proxy

4

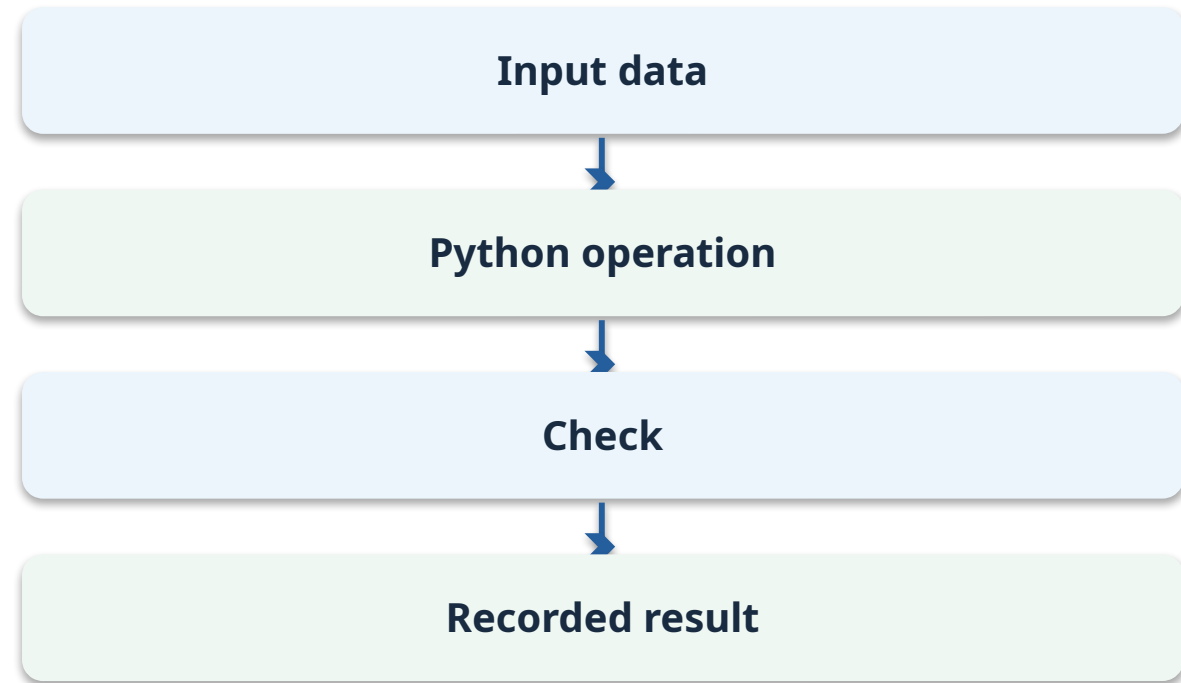
Measure whether the known policy is in the top two

Find the input and check in Python

PRACTICE / WEEK 06

Locate the supplied input, the transformation, the checking line and the printed result.

FOLLOW THE DATA THROUGH EACH ARROW



Predict one controlled change

PRACTICE / WEEK 06

Remove a synonym or lower top-k from 2 to 1 and compare recall.

What should change in the result? What stays fixed?

Before: original case

After: one change

Keep all other inputs fixed.

Run, compare and record

PRACTICE / WEEK 06

Record the input, the one changed setting, the observed result and a reason for any difference.

Input	Setting	Result
Original	baseline	record
Changed	one edit	compare

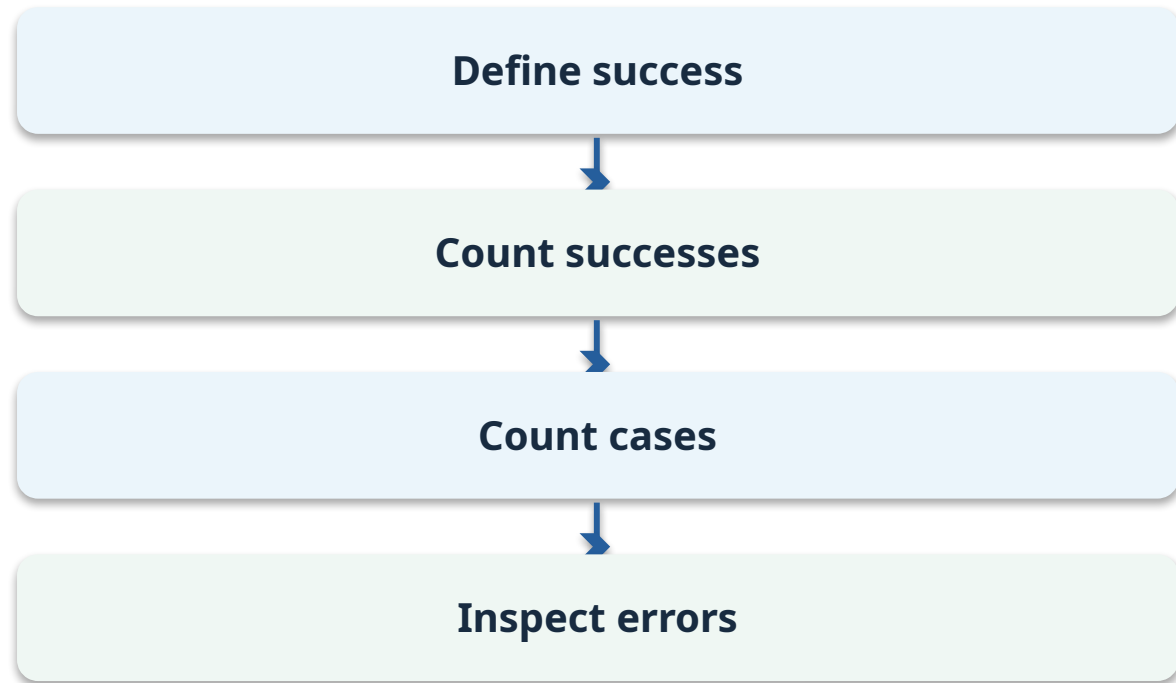
Measure without hiding the denominator

CHECK / WEEK 06

Recall@2 on ten labelled questions

Write the numerator, denominator and one failure case.

FOLLOW THE DATA THROUGH EACH ARROW



Error detective: find the layer

CHECK / WEEK 06

Plausible wrong claim: The synonym mapping is a production embedding model.

Correct it using the case evidence.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

The synonym mapping is a production embedding model.

SUPPORTED EXPLANATION

It is an inspectable proxy for class discussion. A vector index would require a tested embedding model.

Name a test that would catch this failure.

Transfer to a new situation

CHECK / WEEK 06

A staff-only policy has a perfect keyword match. Should the student see it?

Explain which source or test would settle it.

VOTE FIRST • GIVE A REASON

A Use the visible evidence

B Flag missing or unsafe evidence

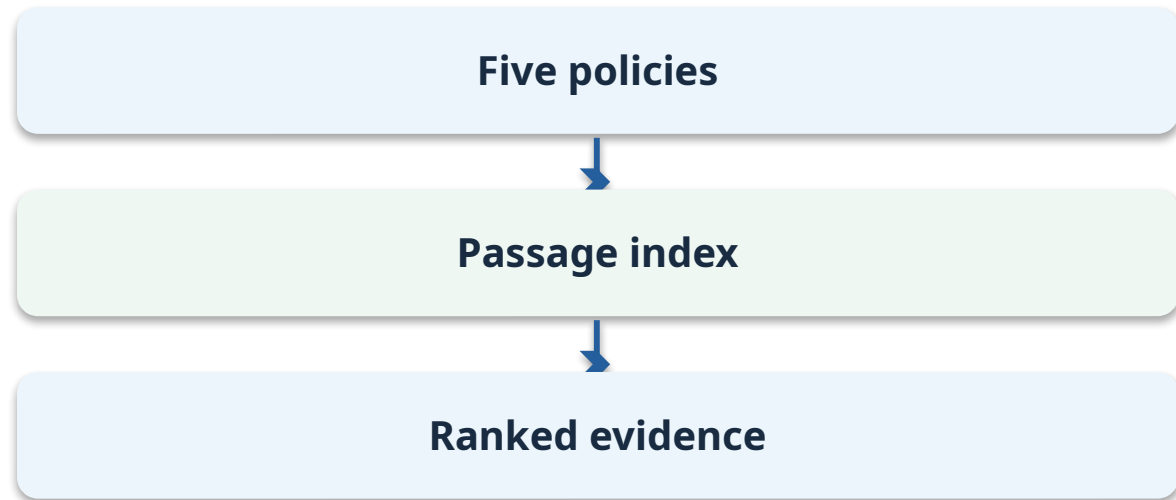
Which source or test could settle this?

Explain it to a partner

CHECK / WEEK 06

Without reading the slide, name the input, one processing step and the check. Your partner should ask “How do you know?”

FOLLOW THE DATA THROUGH EACH ARROW



Exam rehearsal · explain and apply

CHECK / WEEK 06

Why keep source IDs and versions with chunks?

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

Compare with the Word guide

Exam rehearsal · explain and apply

CHECK / WEEK 06

One relevant passage exists and appears in the top two. What is recall@2?

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

Compare with the Word guide

What to hand in

PRACTICE / WEEK 06

Corpus manifest and retrieval misses with evidence IDs.

Include the one changed input and an error explanation.

1

Original run

2

One changed input

3

Comparison

4

Explanation + source

Three ideas to recall next week

CHECK / WEEK 06

Split with source IDs, compare retrieval methods, and filter permissions first.

Read the linked official sources in the Word guide and redraw one diagram from memory.

1

Chunk

2

Keyword search

3

Vector search

4

Access filter